LionX compute: Amazon ECS

LionX is moving to AWS. First decision to lock in: where its services run.

ECS Fargate | ECS on EC2 | EKS

The options › The numbers › Our recommendation › **Your GO**

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The decision, and what drives it

One choice to make: where LionX compute runs on AWS.

THE WORKLOAD

8 services
long-running, plus cron

8 vCPU, 32 GB
per environment

5 environments
integration to production

Networking, CI/CD, observability and IAM all hang off this choice.

WHAT DRIVES THE DECISION

1. **End-2026 cutover.**
   The platform has to fit the migration runway, not the other way around.
2. **A rewrite either way.**
   The deployment layer is rewritten regardless. Today's cluster is a delivery vehicle, not a commitment.
3. **Audit surface stays minimal.**
   Every extra component is more to audit, scan, log and upgrade.
4. **A single workload.**
   No co-tenants to amortize a cluster's overhead against.

Three options

Same workload on each, judged on what costs engineer time.

	ECS Fargate AWS's container orchestrator, serverless: we ship containers, AWS runs the machines	ECS on EC2 the same orchestrator, on EC2 virtual machines that we own and patch	EKS standard to Auto Mode managed Kubernetes: AWS hosts the control plane, we operate the cluster
KUBERNETES TO RUN	none	none	cluster, upgrades, GitOps
NODE OPERATIONS	AWS's job	ours: AMIs, patching, CVEs	ours, or AWS's with Auto Mode
AUDIT SURFACE	CloudTrail only	plus OS-level evidence	plus Kubernetes audit logs
ECOSYSTEM	cron via EventBridge, no Helm	cron via EventBridge, no Helm	CronJob, Helm, operators
FIT FOR LIONX	runs it all, nothing unused	runs everything, ops attached	richer ecosystem, unused

The cheapest bill buys the most ops. Numbers on the next slide, full pros and cons in the ADR.

The numbers

Yearly, for the LionX footprint. Zurich list prices with a 1-year Savings Plan, ops at a conservative $1,000 a day.

	ECS Fargate	ECS on EC2	EKS standard to Auto Mode
INFRA, PER YEAR	~$4.6K	~$3.1K	~$4.3K to 5.2K
OPS, HOURS PER MONTH	~2 to 4	~10 to 20	~15 to 60
OPS, PER YEAR	~$3K to 6K	~$15K to 30K	~$23K to 90K
FULLY LOADED, PER YEAR	~$8K to 11K	~$18K to 33K	~$27K to 95K

Fully loaded, Fargate is the cheapest option, not the priciest. Ops hours decide it.

Infra: eu-central-2 list prices, AWS Pricing API, July 2026. Ops: assumptions built from Encore, CloudAtler, CloudOptimo and Spectro Cloud. Not measurements. Sourced and detailed per EKS variant in the ADR.

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Decision: ECS Fargate

The smallest operational surface that runs all of LionX.

ECS Fargate

all eight services and the scheduled tasks, one runtime, zero nodes

Zero Kubernetes to run
no cluster, no CNI, no GitOps reconciler, no upgrade calendar

One audit surface
CloudTrail end to end, host patching is AWS's responsibility

Nothing blocks landing
no cluster standup on the critical path to end 2026

Accepted trade-offs. Each schedule is an EventBridge rule plus an IAM role, more Terraform than a CronJob. The ecosystem is narrower than Kubernetes, and the difference would sit unused.

Next steps

What we ask of you today.

Questions and objections

here and now, let's settle them in the room

Ratify today

the ADR moves from proposed to accepted

PoC

pipeline, IAM boundaries and the cron path, end to end in the new AWS account

The ask today: your GO on ECS Fargate.

Any questions? Let's settle them now and call it.